



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Scalar and Predictive Quantization

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Multimedia Communications



Definitions

Uniform Quantization

Examples

Rate-distortion curve

Optimal Scalar Quantization

Predictive scalar quantization



Definitions

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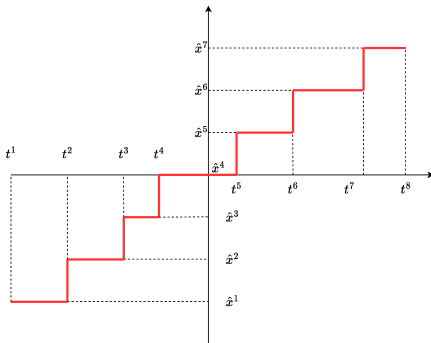


Scalar quantization (SQ): it is just a staircase function (almost always)

Here we show the graph of a possible $Q(x)$

Go to [wooclap.com](https://www.wooclap.com) with the code **PBTFUO**

Input range:





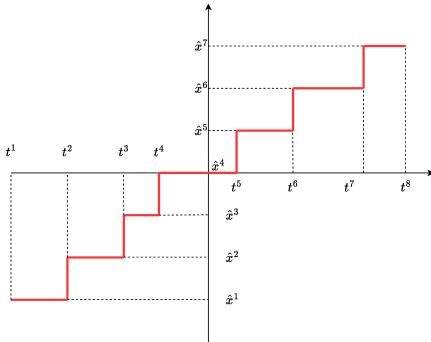
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Input range: (t_1, t_8)

Number of levels:





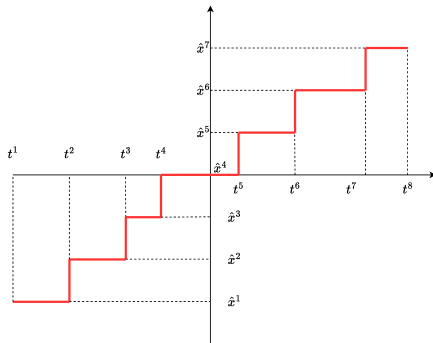
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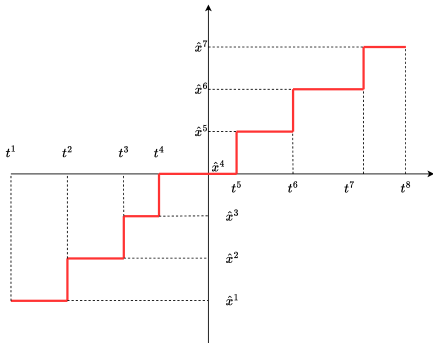
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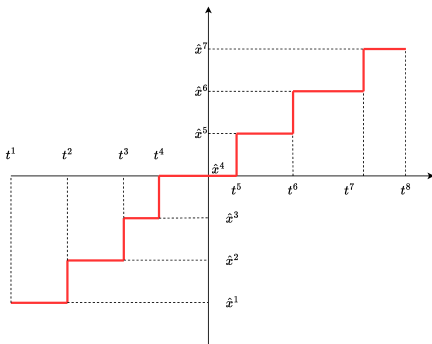
The t^i 's determine a *partition* of the input interval

Each $\Theta^j = (t^j, t^{j+1})$ is a quantization region or cell

Any number in $\Theta^i = (t^i, t^{i+1})$ is quantized as $\hat{x}^i$



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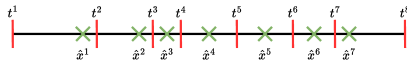
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The graph below also represent the quantizer: the thresholds and the levels completely determine the quantizer





Definition : scalar quantization (SQ)

Scalar quantization (SQ) is a function Q from $\mathbb{R}$ to a discrete set called *Dictionary*

$$Q : x \in \mathbb{R} \rightarrow y \in \mathcal{C} = \{\hat{x}^1, \hat{x}^2, \dots, \hat{x}^L\} \subset \mathbb{R}$$

- ▶ $\mathcal{C}$: Dictionary, it is a discrete subset of $\mathbb{R}$
- ▶ $\hat{x}^i$: quantization level, code-word
- ▶ $e = x - Q(x)$: Quantization noise
- ▶ $\Theta^i = \{x : Q(x) = \hat{x}^i\}$: Decision regions or cells

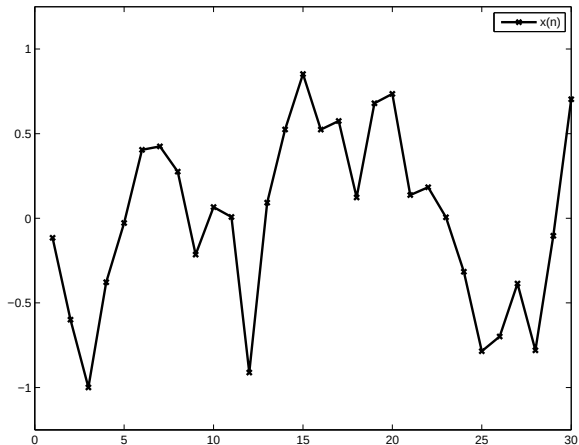
Regions and levels completely define the SQ

Regions are typically defined as intervals: $\Theta^i = (t^i, t^{i+1})$

Therefore, thresholds and levels also completely define a SQ

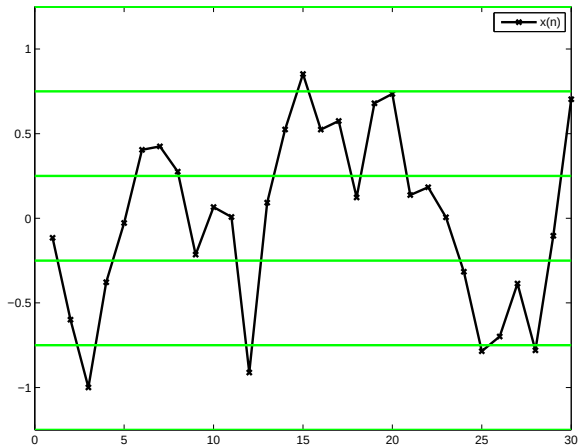


Example 1





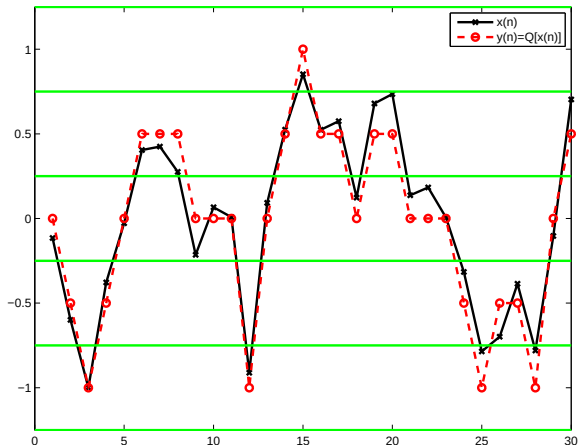
Example 1







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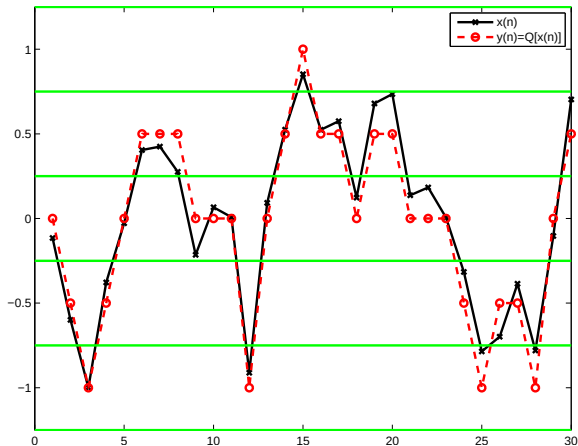
Go to wooclap.com with the code **PBTFOU**

Input interval: $(-A, A) = (-1.25, 1.25)$

Thresholds:



Example 1



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Input interval: $(-A, A) = (-1.25, 1.25)$

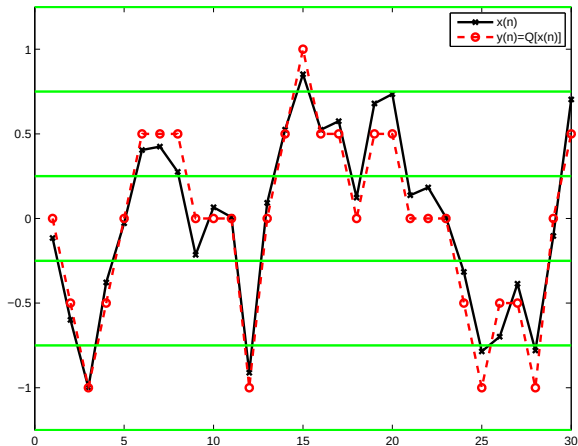
Thresholds: $-1.25, -0.75, -0.25, 0.25, 0.75, 1.25$







Example 1



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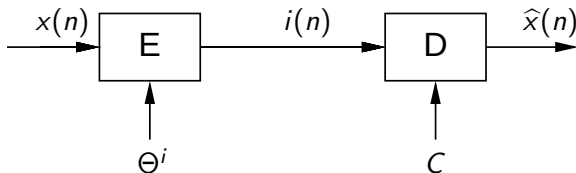
Thresholds: $-1.25, -0.75, -0.25, 0.25, 0.75, 1.25$

Levels: $-1, -0.5, 0, 0.5, 1$

Note that:

- ▶ All the cells have the same size
 $\Delta = \frac{2A}{L} = 2.5/5 = 0.5$
- ▶ Each level is the center of the corresponding cell





- ▶ The quantization process can be seen as the cascade of “encoding” and “decoding”
- ▶ The encoder produces the index of the quantization region, $i(n) \in \mathbb{Z}$
- ▶ The number $i(n)$ is represented with a suitable string of bits, e.g by using *lossless coding*
- ▶ The decoder associates to $i(n)$ to the corresponding level (or code-word) $\hat{x}^i(n)$
- ▶ In a compression system, often with the term quantization we just refer to the operation $x \rightarrow i$, while the evaluation of the quantization level, $i \rightarrow \hat{x}^i$ is referred to as *Inverse Quantization*
 - ▶ The first is performed at the encoder, the second at the decoder



Rate of a scalar quantizer

- ▶ **Rate** R and **distortion** D are the performance criteria of a quantizer
- ▶ The rate of a scalar quantizer is the average number of bits needed to represent a sample, or rather, its quantization index: it amounts thus to the binarization of $i(n)$
- ▶ It is *always done via lossless coding*
- ▶ We will assume $R = \log_2 L$
- ▶ This is a good approximation of real-life coding rate, with two implicit assumption



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- ▶ This is a good approximation of real-life coding rate, with two implicit assumption
 - ▶ Quantized data have a uniform probability distribution (pessimistic)
 - ▶ The binary representation of $i(n)$ is the best (entropy coder)
- ▶ We will elaborate on this problem in the next lesson



Distortion of a scalar quantizer

- We use the squared error, which, on a single sample is:

$$d[x(n), Q[x(n)]] = |e(n)|^2 = |x(n) - Q[x(n)]|^2$$

- For a signal $x(\cdot)$ of duration N , we use the mean square error (MSE):

$$D = \frac{1}{N} \sum_{n=0}^{N-1} d[x(n), Q[x(n)]]$$

- ▶ For random signals¹, $D = E \left\{ |X(n) - Q(X(n))|^2 \right\} = E \left\{ |E(n)|^2 \right\}$
- ▶ In this case, distortion is the variance of the random process $E(n) = X(n) - Q(X(n))$, and is indicated as σ_Q^2

¹Notice that we use lower-case letters for deterministic signals and upper-case letter for random variables/signals, while $\mathbb{E}$ is the expectation operator.



Mathematical reminders: integer mapping

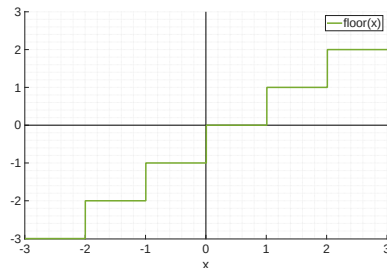
- ▶ Let us remind the integer mapping functions commonly used in programming:
floor, ceil, round, fix
- ▶ They are “staircase functions”
- ▶ They are widely used to implement quantizers



Mathematical Reminders: Floor Function

Operator: $|x|$

- ▶ **Definition:** The greatest integer less than or equal to x
- ▶ **Behavior:** Always rounds towards $-\infty$
- ▶ **Examples:**
 - ▶ $\lfloor 2.7 \rfloor = 2$
 - ▶ $\lfloor -2.7 \rfloor = -3$
- ▶ **Usage:** Common in basic uniform quantization models

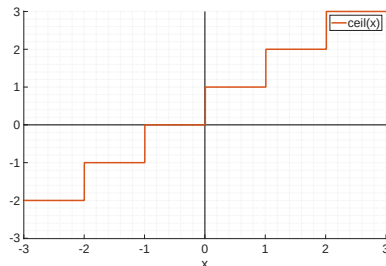




Mathematical Reminders: Ceil Function

Operator: $\lceil x \rceil$

- ▶ **Definition:** The smallest integer greater than or equal to x
- ▶ **Behavior:** Always rounds towards $+\infty$
- ▶ **Examples:**
 - ▶ $\lceil 2.1 \rceil = 3$
 - ▶ $\lceil -2.1 \rceil = -2$
- ▶ **Usage:** Used for determining buffer sizes or resource allocation bounds

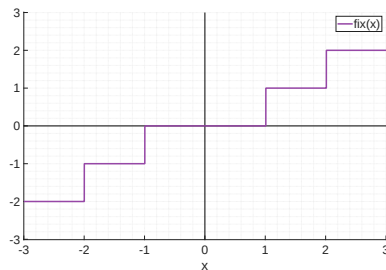




Mathematical Reminders: Fix Function

Operator: $\text{fix}(x)$

- ▶ **Definition:** Drops the fractional part of the number
- ▶ **Behavior:** Rounds towards zero (Symmetric)
 - ▶ Equivalent to $\lfloor x \rfloor$ for $x \geq 0$.
 - ▶ Equivalent to $\lceil x \rceil$ for $x < 0$.
- ▶ **Examples:**
 - ▶ $\text{fix}(2.7) = 2$
 - ▶ $\text{fix}(-2.7) = -2$
- ▶ **Note:** Standard behavior for integer casting in most programming languages

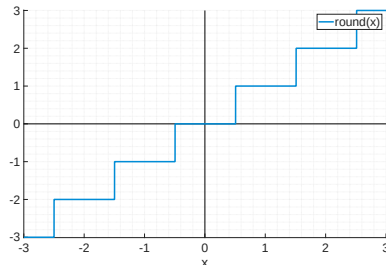




Mathematical Reminders: Round Function

Operator: $\text{round}(x)$

- ▶ **Definition:** Maps x to the nearest integer
- ▶ **Behavior:** Usually implemented as $\lfloor x + 0.5 \rfloor$
- ▶ **Examples:**
 - ▶ $\text{round}(2.51) = 3$
 - ▶ $\text{round}(2.49) = 2$
 - ▶ $\text{round}(-2.3) = -2$
- ▶ **Usage:** allow a straightforward implementation of *uniform quantization*





Definitions

Uniform Quantization

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Rate-distortion curve

Optimal Scalar Quantization

Predictive scalar quantization



Uniform quantization

Uniform quantization is characterized by:

- ▶ The input range: $(0, A)$ (unsigned data) or $(-\frac{A}{2}, \frac{A}{2})$ (signed data)
- ▶ The number of levels L

The input range is divided into L equal-sized cells. The cell size is $\Delta = \frac{A}{L}$. Each cell is represented by its mid-point, i.e. the quantization levels are the centers of the quantization cells.

$$\forall i, \Delta^i = \Delta = A/L$$

$$t^i = t^{i-1} + \Delta \text{ (red bars)}$$

$$\hat{x}^i = \frac{t^i + t^{i-1}}{2} \text{ (green crosses)}$$

$$\Theta^i = \left(\hat{x}^i - \frac{\Delta}{2}, \hat{x}^i + \frac{\Delta}{2} \right)$$





Uniform Quantization for unsigned data

For **unsigned data** (range $(0, A)$) the thresholds are $(0, \Delta, 2\Delta, \dots, L\Delta)$, i.e., $t^i = (i-1)\Delta$.

We can thus define the behavior of the **encoder** (from x to the quantization index i), of the **decoder** (from i to the quantization level $\hat{x}^i$) and of the full quantization chain:

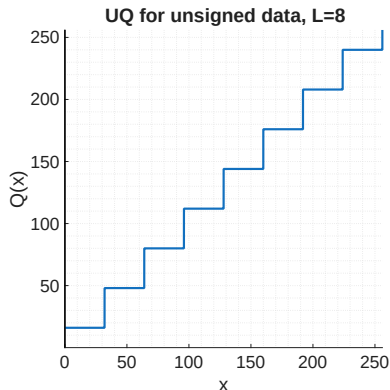
$$i = \left\lceil \frac{x}{\Delta} \right\rceil$$

encoder

$$\hat{x}^i = i \cdot \Delta - \frac{\Delta}{2}$$

decoder

$$Q(x) = \Delta \cdot \left\lceil \frac{x}{\Delta} \right\rceil - \frac{\Delta}{2}$$





Uniform quantization for signed data

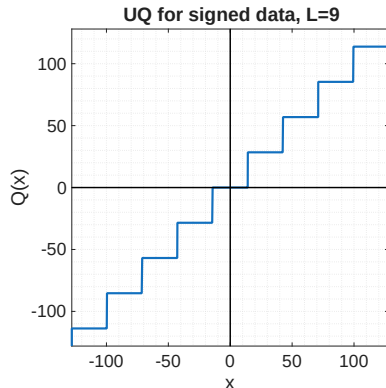
For **signed data** (range in $(-A/2, A/2)$) one typically wants zero to be a quantization level (**mid-tread quantizer**): in such a way, small fluctuations are quantized to zero. Therefore, we **always use an odd number of levels for UQ of signed data**. In this case, uniform quantization (UQ) amounts to **rounding**:

$$i = \text{round} \left(\frac{x}{\Delta} \right) \quad \text{encoder}$$

$$\hat{x}^i = \Delta \cdot i \quad \text{decoder}$$

$$Q(x) = \Delta \cdot \text{round} \left(\frac{x}{\Delta} \right)$$

In other words, $Q(x)$ is **the closest multiple of Δ** . This quantizer is also referred to as **mid-tread quantizer**.



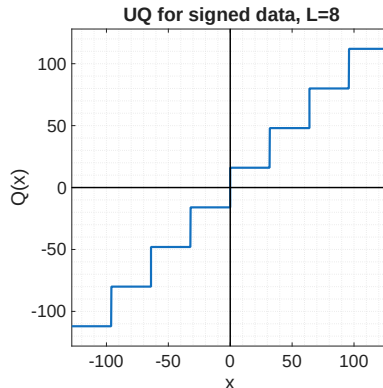


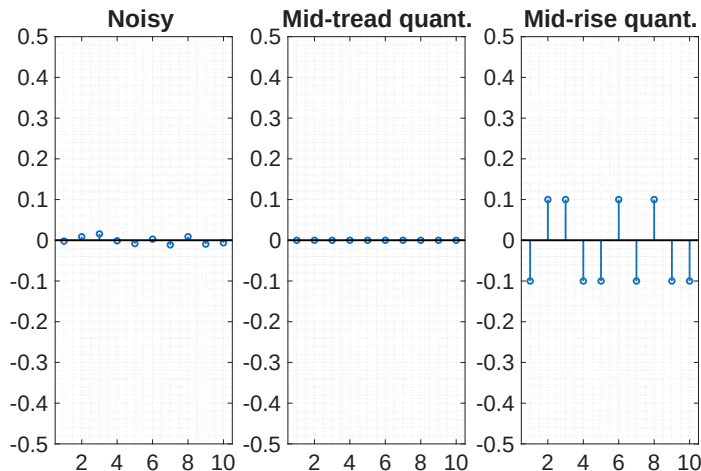
Uniform quantization for signed data

If one uses uniform quantization with an even number of levels for signed data, the central threshold $t^{\frac{N}{2}}$ is zero, see figure on the right.

This quantizer is also referred to as **mid-rise** quantizer. Since zero is a threshold any very small value will be quantized as $\pm\Delta/2$ according to its sign.

This can be a problem if the signal has small fluctuations around zero due to additive noise: quantization enhances noise!





On the left, we show $x(n)$, a signal with small values that are **interpreted** as noise: uninformative, random fluctuations.

In the middle, we show $Q(x)$ when zero is a quantization level (**mid-tread** quantizer): the small values are forced to zero, making it “easier” to encode the signal.

On the right, we show $Q(x)$ when zero is a threshold (**mid-rise** quantizer): after quantization the noise is amplified instead of being removed!



UQ: wrap-up

- ▶ UQ is intuitive: just split the input range into L intervals, and represent each interval with its middle.
- ▶ Implementation is easy:

$$Q(x) = \Delta \cdot \text{round}\left(\frac{x}{\Delta}\right) \quad \text{mid-tread}$$

$$Q(x) = \Delta \cdot \text{floor}\left(\frac{x}{\Delta}\right) + \frac{\Delta}{2} \quad \text{mid-rise}$$

- ▶ For a given range A and a given number of levels L , UQ has the smallest maximum error (optimal **minimax** quantizer)
 - ▶ The maximum error is half the size of the maximum quantization interval: any modification of the UQ increases the maximum quantization interval and thus the maximum error
- ▶ Last, but not least, UQ allows for an analytical formula of the RD (rate-distortion) curve. But before that...



Def

Unit

Exa

Rate

Opt

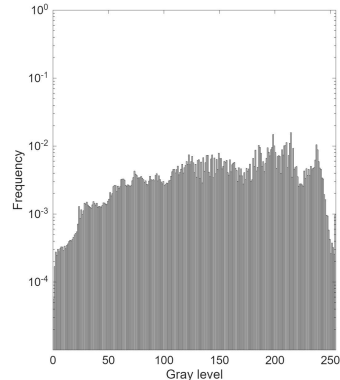
Pre





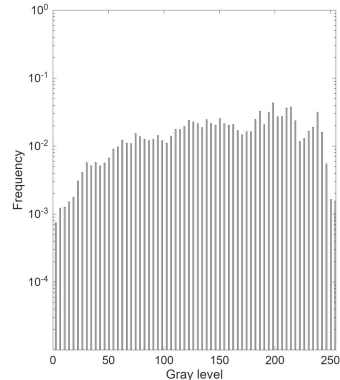


Scalar quantization: example 2



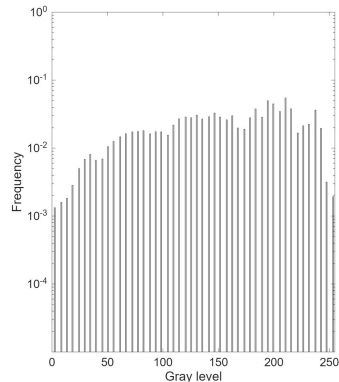


Scalar quantization: example 2



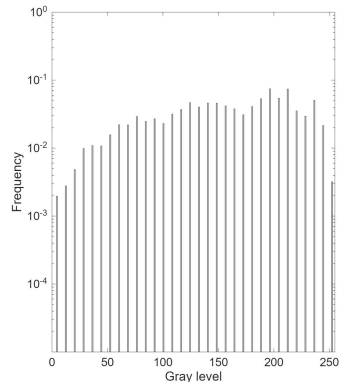


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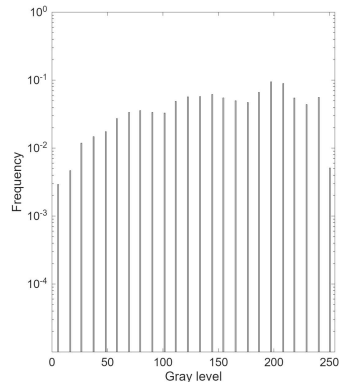


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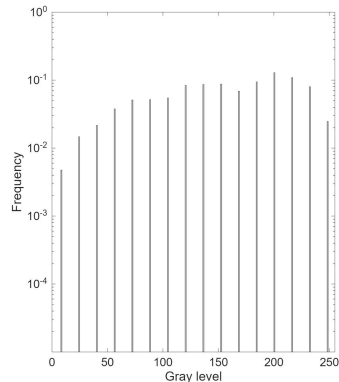


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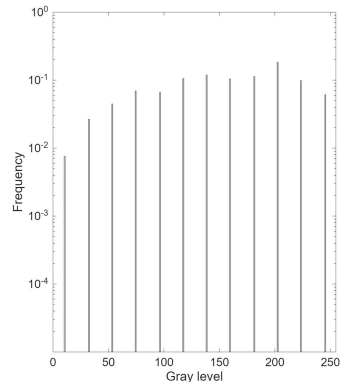


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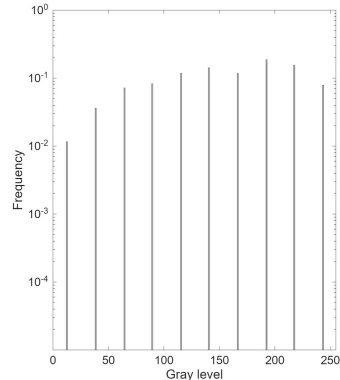


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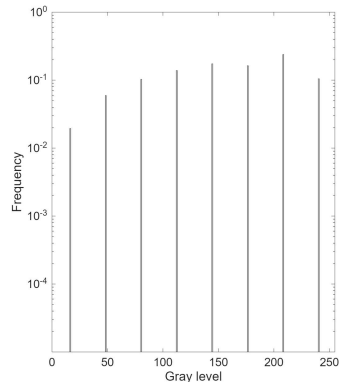


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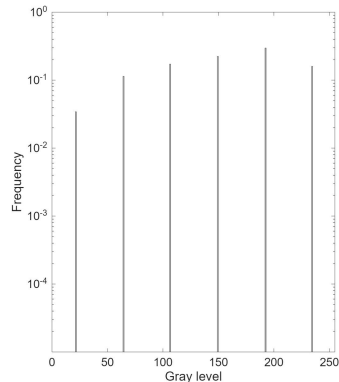


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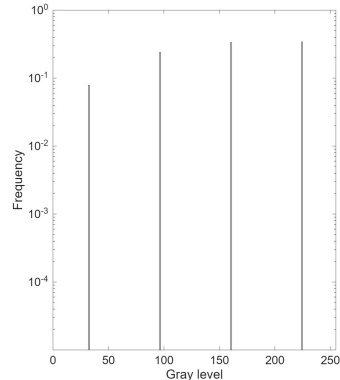


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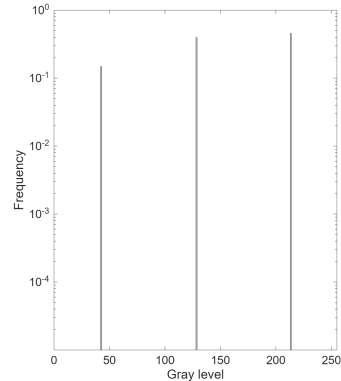


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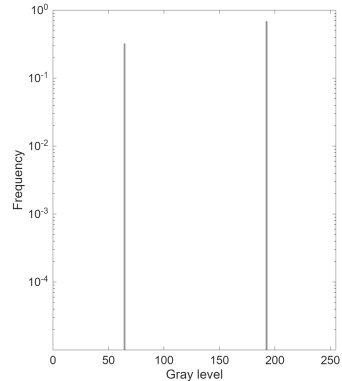
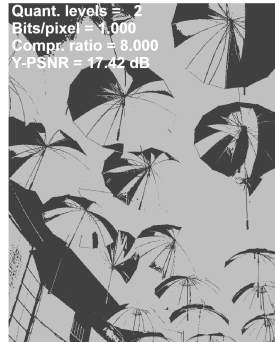


Scalar quantization: example 2





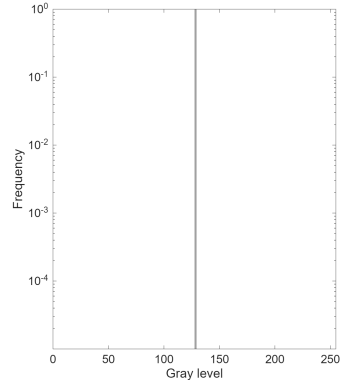
Scalar quantization: example 2





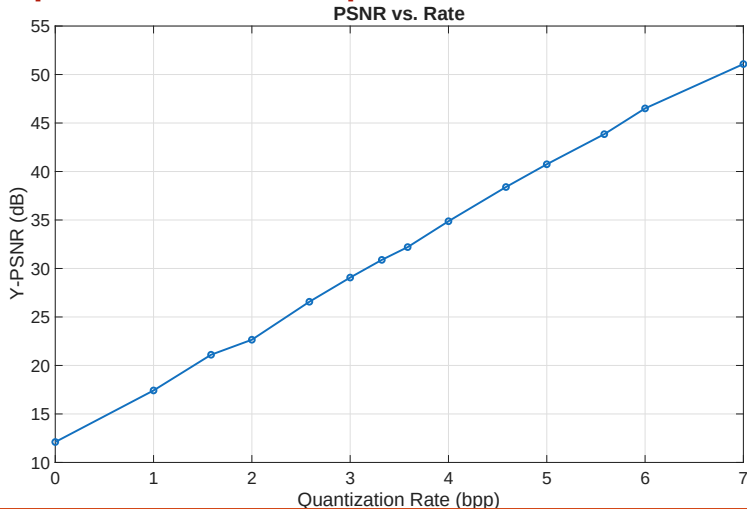
Scalar quantization: example 2

Quant. levels = 1
Bits/pixel = 0.000
Compr. ratio = Inf
Y-PSNR = 12.11 dB





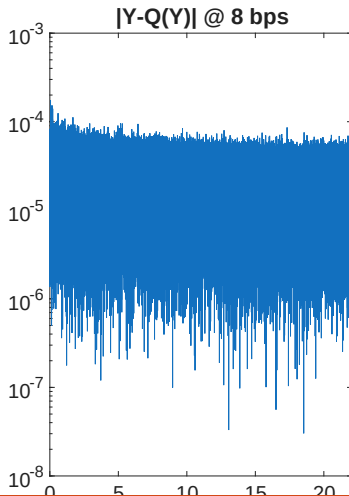
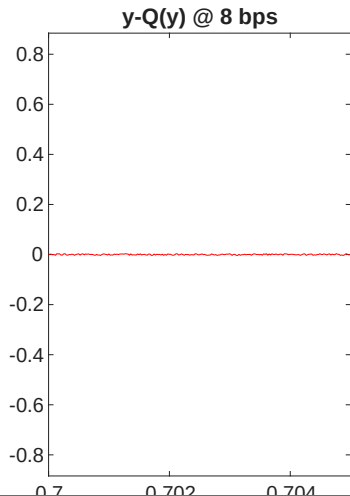
Scalar quantization: example 2





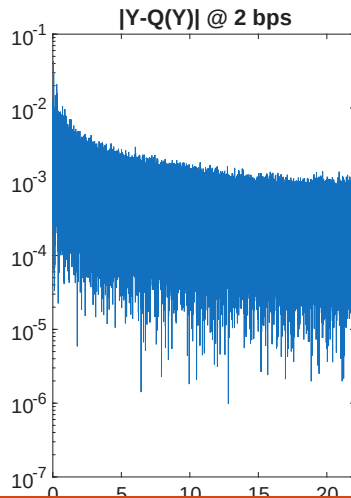
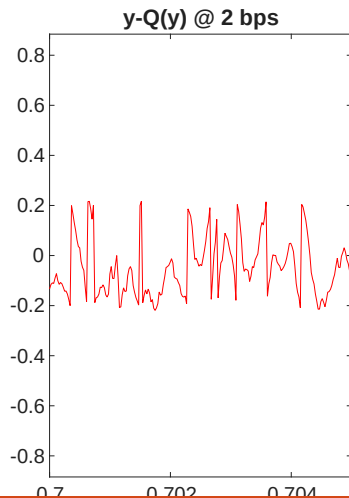


Audio quantization example





Audio quantization example







Outline

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